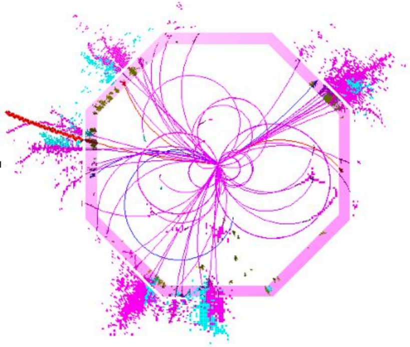


DEEP UNDERGROUND NEUTRINO EXPERIMENT

DUNE Columnar Analysis

Alex, Layne, Lauren
Baryons and Cream

Northern Illinois University

Overview

- DUNE
- ProtoDUNE
- DUNE Particle Physics
- Pandora
- Particle Identification
- Columnar Analysis
- Results

Neutrino 101

- Neutrinos are emitted and absorbed in flavor eigenstates
- Neutrinos travel as mass eigenstates (m_1, m_2, m_3)
  - A single mass does not correspond to a single flavor rather each flavor has a different mass mixture
- The mass eigenstates are a superposition of each of the masses of the different neutrino flavors
- The mixing that occurs while traveling is called **oscillation**.
- The probability of these oscillations are determined partially by the Pontecorvo-Maki-Nagakawa-Sakata (PMNS) Matrix
- Parameters of the PMNS Matrix are θ_{12}, θ_{13}, θ_{23}, Δm_{12}, Δm_{23}, δ_{CP}

PMNS Matrix

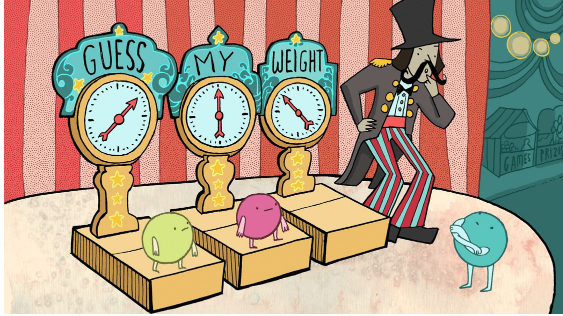
$$
\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix}
\begin{bmatrix} c_{13} & 0 & s_{13} e^{-i\delta_{\mathrm{CP}}} \\ 0 & 1 & 0 \\ -s_{13} e^{i\delta_{\mathrm{CP}}} & 0 & c_{13} \end{bmatrix}
\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

$$
= \begin{bmatrix} c_{12} c_{13} & s_{12} c_{13} & s_{13} e^{-i\delta_{\mathrm{CP}}} \\ -s_{12} c_{23} - c_{12} s_{23} s_{13} e^{i\delta_{\mathrm{CP}}} & c_{12} c_{23} - s_{12} s_{23} s_{13} e^{i\delta_{\mathrm{CP}}} & s_{23} c_{13} \\ s_{12} s_{23} - c_{12} c_{23} s_{13} e^{i\delta_{\mathrm{CP}}} & -c_{12} s_{23} - s_{12} c_{23} s_{13} e^{i\delta_{\mathrm{CP}}} & c_{23} c_{13} \end{bmatrix},
$$

DUNE Goals

1. Determine Neutrino Mass Hierarchy
2. Measure δ_{CP}, θ_{23}, and other PMNS parameters
3. Detect low energy neutrinos from supernovae
4. Find the White Whales (nucleon decay, sterile/heavy neutrinos, neutrino tridents, dark matter, other BTSM stuff)

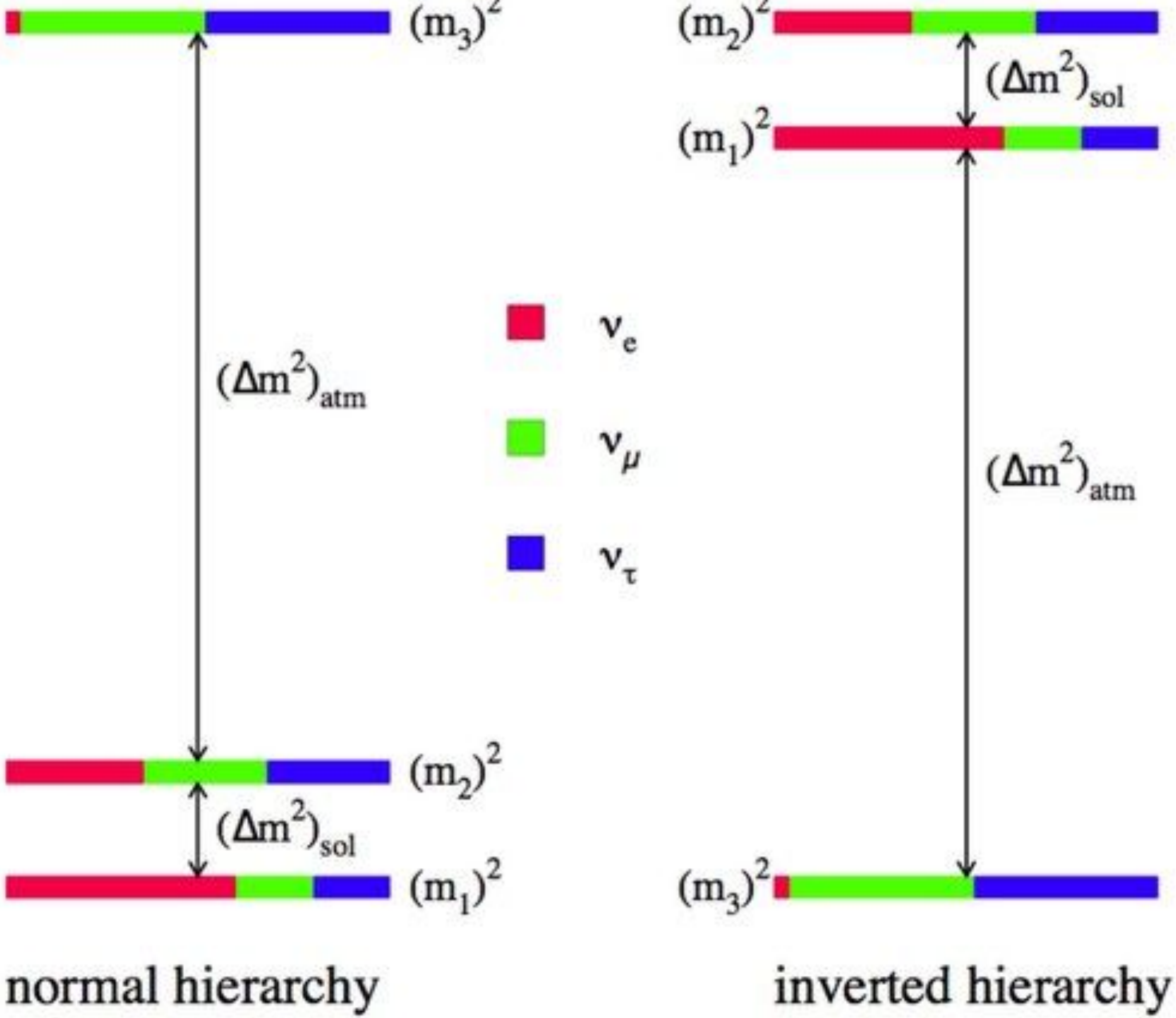
Neutrino Mass Hierarchy



- The probability of an oscillation is a function of $(\Delta m)^2$
- Since this probability relies on $(\Delta m)^2$, the sign of Δm needs to be determined to know the hierarchy
- There are two possibilities
  - Normal Ordering: $m_3 > m_2 > m_1$
  - Inverse Ordering: $m_2 > m_1 > m_3$
- Using the antineutrino channel and the asymmetry of electrons to positrons in the Earth, the sign of Δm_{13} may be determined (and thus the mass hierarchy)

$$P_{\alpha\to\beta} = \delta_{\alpha\beta} - 4\sum_{j>k} \mathcal{R}_e\left\{U^*_{\alpha j}\,U_{\beta j}\,U_{\alpha k}\,U^*_{\beta k}\right\}\sin^2\left(\frac{\Delta_{jk}m^2\,L}{4E}\right)$$
$$+\,2\sum_{j>k} \mathcal{I}_m\left\{U^*_{\alpha j}\,U_{\beta j}\,U_{\alpha k}\,U^*_{\beta k}\right\}\sin\left(\frac{\Delta_{jk}m^2\,L}{2E}\right),$$

where $\Delta_{jk}m^2 \equiv m_j^2 - m_k^2$.


$(m_3)^2$
$(\Delta m^2)_{atm}$
$(m_2)^2$
$(\Delta m^2)_{sol}$
$(m_1)^2$
normal hierarchy
ν_e
ν_μ
ν_τ
$(m_2)^2$
$(\Delta m^2)_{sol}$
$(m_1)^2$
$(\Delta m^2)_{atm}$
$(m_3)^2$
inverted hierarchy

PMNS Matrix parameters

- δ_{CP}
  - Measuring oscillation parameters for neutrinos and antineutrinos would allow us to measure δ_{CP} for both
  - If neutrinos and antineutrinos oscillate differently, then there should be CP violation
- θ_{23} octant
  - Atmospheric mixing angle
  - Experimentally shown to be not the maximum of 45°
  - Parameter measured as $\sin^2 \theta_{23}$ so the true value could be above or below 45°
  - Off axis measurements plus neutrino-antineutrino measurements should help determined this angle
- Other Parameters
  - The rest of the PMNS parameters should be able to find tighter constraints for each

Low Energy Neutrinos

- Sensitive to neutrinos in range of 5-100 MeV
- FD connected to SuperNovae Early Warning Systems (SNEWS)
- Low energy sensitivity combined with the four independent modules makes DUNE great for this task

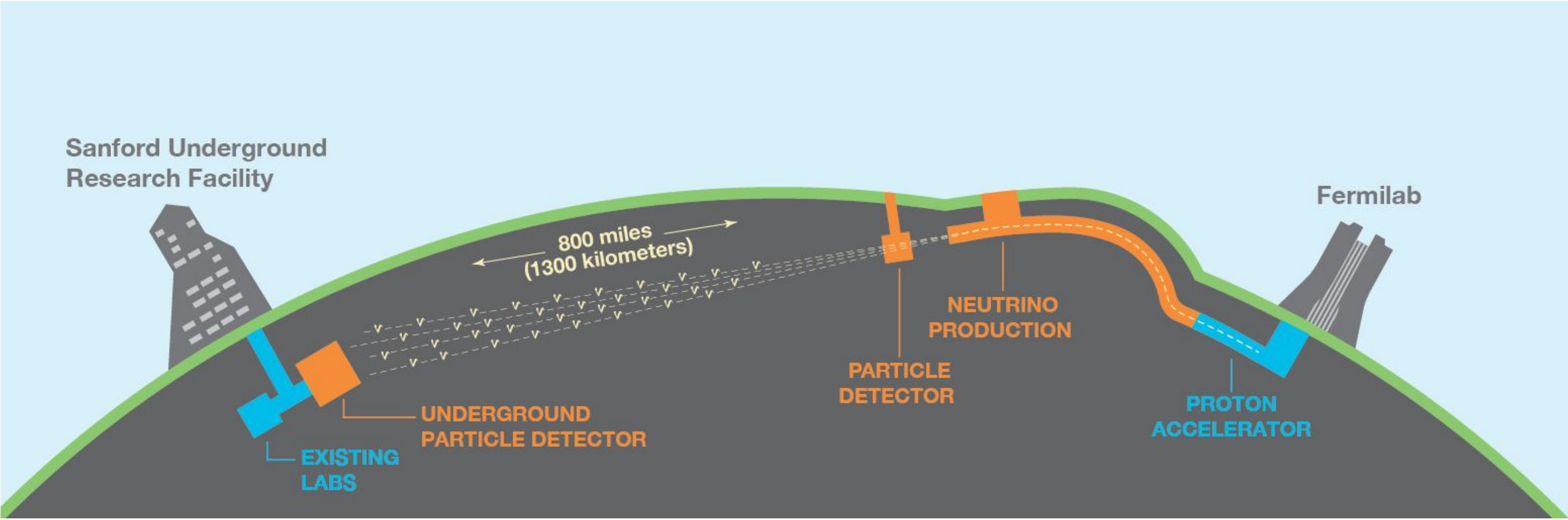
White Whales

- Proton Decay
  - DUNE has a wide range of energy sensitivity
  - Proton Decay products would fall within this range and be detected by DUNE
- Sterile Neutrino(s)
  - Determination of delta CP could lead to possibility of other neutrinos
- Tridents
  - Neutrino interactions off heavy nuclei that produce lepton/antilepton pairs
- Dark Matter
  - Could encompass numerous effects that would fall into the detection range of DUNE

DUNE Structure

- Neutrino Beam
  - At Fermilab
  - Made from a Proton beam hitting a graphite target
- Near Detector (ND)
  - At Fermilab
  - Consists of Liquid Argon Time Projection Chamber (LArTPC), The Muon Spectrometer (TMS), System for on-Axis Neutrino Detector (SAND), and Precision Reaction-Independent Spectrum Measurement (PRISM)
- Far Detector (FD)
  - At Sanford Underground Research Facility (SURF) in South Dakota
  - 1300 km from Fermilab
  - Consists of two (Phase I) then four (Phase II) LArTPC's with a total of 35 (70) kton of liquid argon

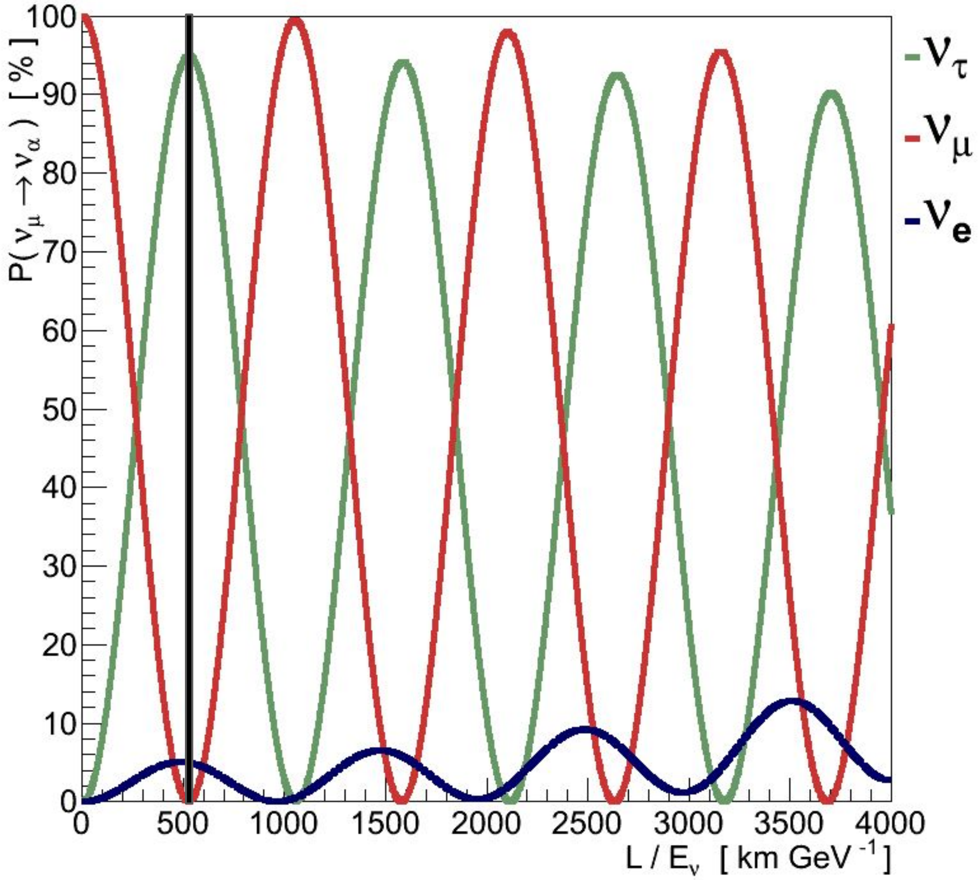
DUNE



Sanford Underground Research Facility
Fermilab
800 miles (1300 kilometers)
NEUTRINO PRODUCTION
PARTICLE DETECTOR
UNDERGROUND PARTICLE DETECTOR
EXISTING LABS
PROTON ACCELERATOR

Why 1300 km?

- L/E peaks at 500 km/GeV
- Most neutrinos will oscillate
    - Mostly tau, some electron/muon



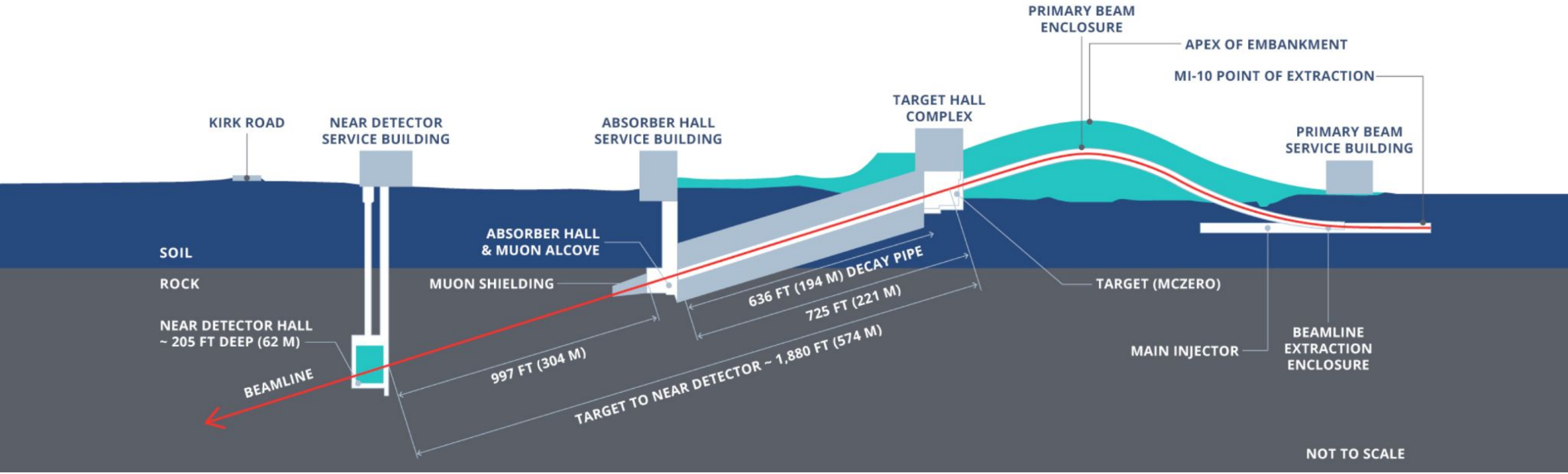
Neutrino Beam

- Accelerated proton beam strikes a graphite target
  - ~85% interact to produce secondary particles
- Kaons and pions are chosen and focused using 2 magnetized horns
  - Chosen for either positive (neutrinos) or negative (antineutrinos)
- In a 194 m length by 4m diameter He-filled chamber the kaons and pions will decay into muons and muon neutrinos
  - Optimized for maximum kaon and pion decay
- Absorber (aluminum, steel, concrete) collects undecayed kaons, pions, and non interacting protons
- The energy of the proton beam is selected so that the muon neutrinos produced are peaked in the range of 0.5 - 5 GeV

Near Detector

- The near detector is 574 m from the neutrino production point
- LArTPC
  - Single Phase/Horizontal Drift (Alternating anodes and cathodes ACACA)
    - Uniform 500 V/cm electric field
    - Charged particles in volume ionize argon, ionized charged particles drift horizontally
    - Charged particles also produce scintillation (photons)
- TMS
  - Neutrino beam is mostly muon neutrinos so interactions will produce muons
- SAND
  - Monitors neutrino beam
  - Measures neutrino cross sections
- PRISM
  - Moves LArTPC and TMS off axis
  - Allows data collection of different fluxes
  - Constrains the energy dependence of neutrino cross sections

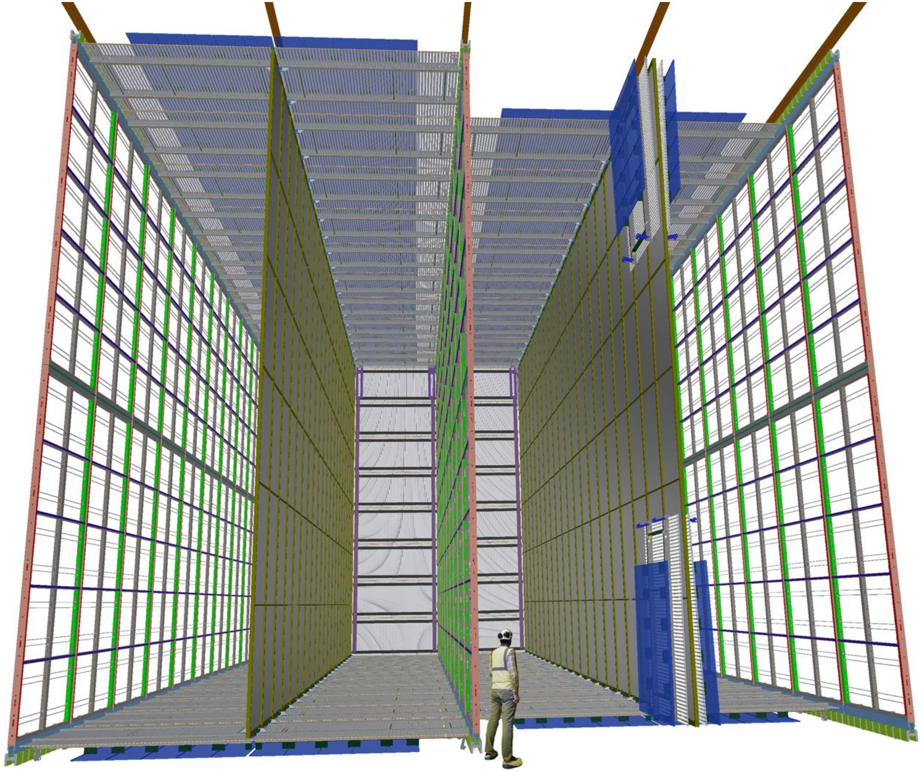
Fermilab Facilities

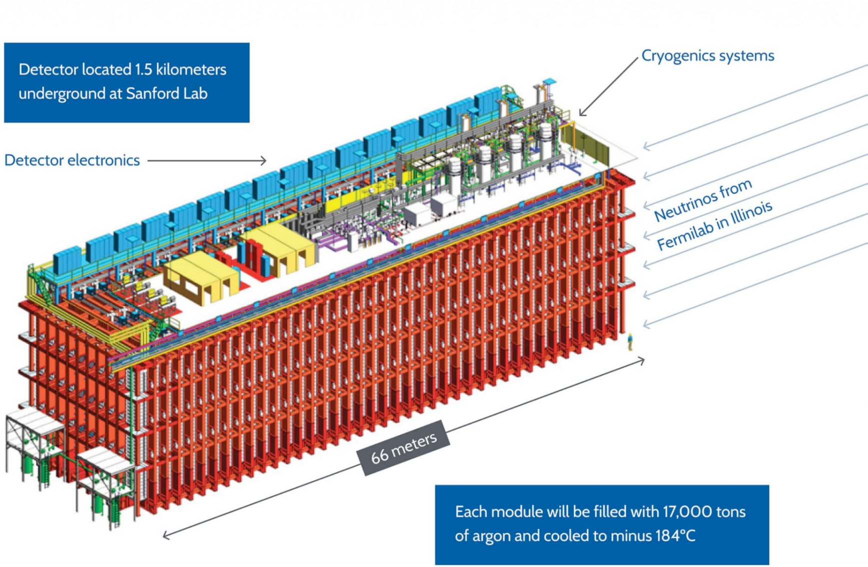


Far Detector

- 1300 km from the neutrino production point
- Four LArTPC with a total volume of 70 kton of liquid argon but fiducial volume of 40 kton
- Two LArTPC will use Horizontal Drift Technology
    - Horizontal Drift (Alternating anodes and cathodes side to side)
        - Uniform 500 V/cm electric field
        - Charged particles in volume ionize argon, ionized charged particles drift horizontally
        - Charged particles also produce scintillation (photons)
- One LArTPC will use Vertical Drift Technology
    - Vertical Drift (anode top, cathode bottom)
        - Easier to construct
        - Employs gaseous Argon as well as liquid
        - Gasous Argon area uses a stronger E field to extract electrons into Large Electron Multipliers
- Final LArTPC is undecided (as of 2020 report I read)
- LArTPC's also have photon detectors
- Large range of sensitivities (low end ~5MeV, high end multiple GeV)

FD at SURF





Detector located 1.5 kilometers underground at Sanford Lab
Cryogenics systems
Detector electronics
Neutrinos from Fermilab in Illinois
66 meters
Each module will be filled with 17,000 tons of argon and cooled to minus 184°C

ProtoDUNE

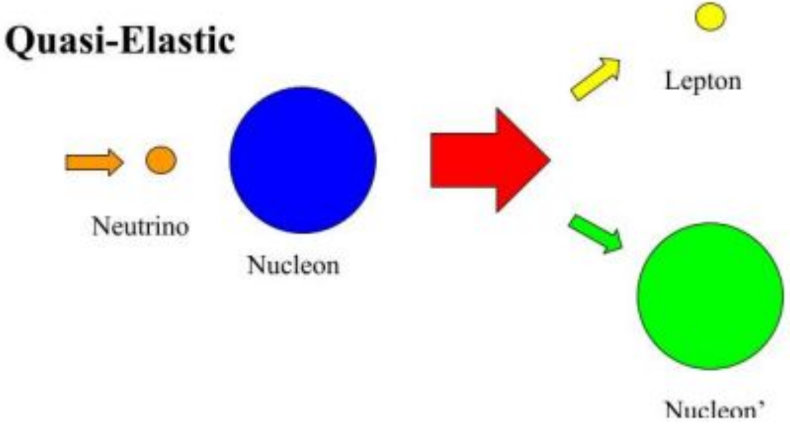
- Neutrino beam of range 0.5 GeV to 7 GeV
- Prototypes of FD modules
  - ProtoDUNE-SP is a 700 ton prototype of the Horizontal Drift module type
  - ProtoDUNE-DP is a prototype of the Vertical Drift module type
- Used to provide insight into DUNE computing challenges
- ProtoDUNE-SP has only ACA instead of ACACA
  - Ran at 25 Hz for 6 weeks
  - “In total 850 TB of raw test beam data were written, along with 1 PB of commissioning and cosmic data.”
- ProtoDUNE-DP not exposed to beam (as of 2022)
- Simulation and Reconstruction performed by LArSoft in 2019, 2020
- Reconstruction performed by Pandora started in 2021

Neutrino Interactions- Layne

- Neutrinos interact weakly, through either charged-current or neutral-current
  - LAr detectors and other detectors *mostly detect CC events
- CC Nucleus-Neutrino scattering
  - Quasi-Elastic
  - Resonant
  - Deep inelastic
  - 2p-2h
- In DUNE energy region, all CC nucleon scattering channels (except 2p-2h) have similar cross sections

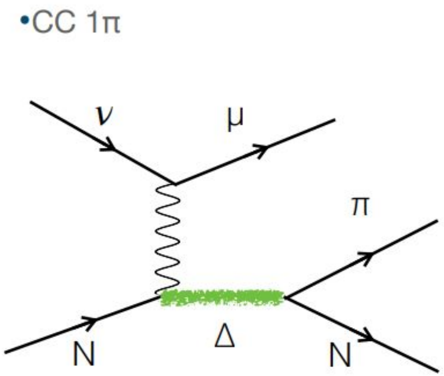
CCQE interaction

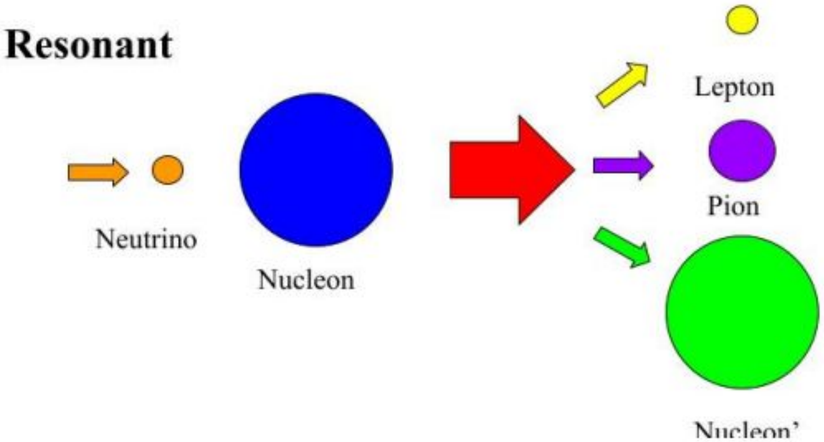
- Simplest CC nucleon-neutrino interaction
- Quasi, final state particles are not the same



Resonant

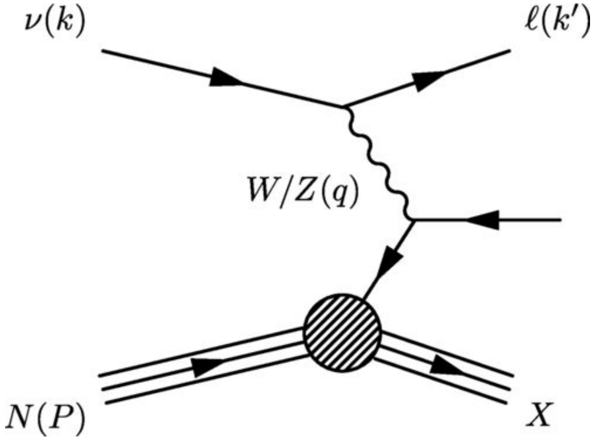
- Higher energy neutrino reaches threshold of resonant particle, like Δ's
- Distinct by usually final state pion
- Pion can be absorbed by nucleus
  - Fake QE event

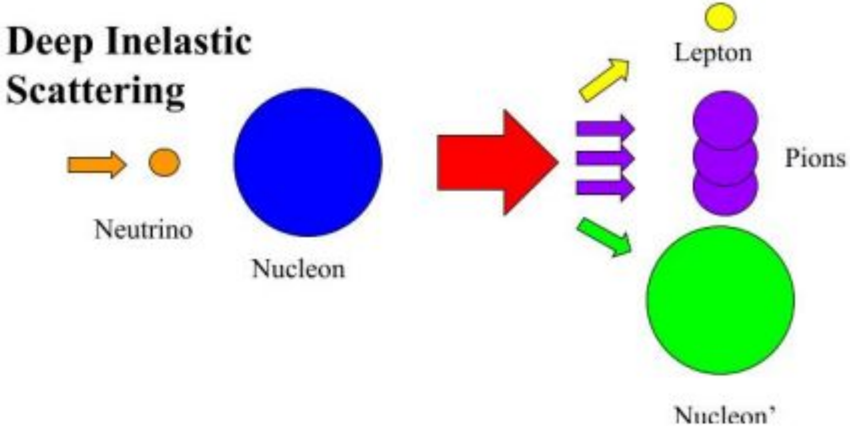




Deep Inelastic

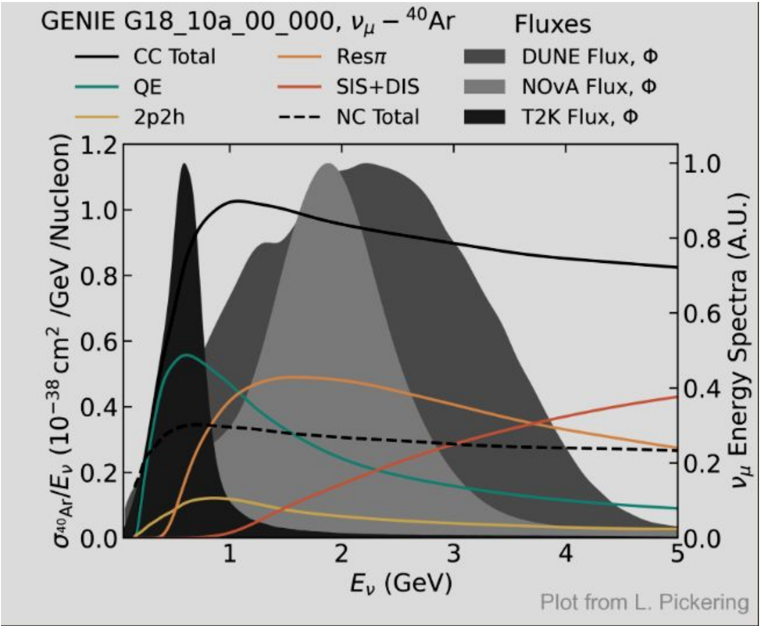
- Easier to identify, many final state particles
- Hadronic showers, usually pions

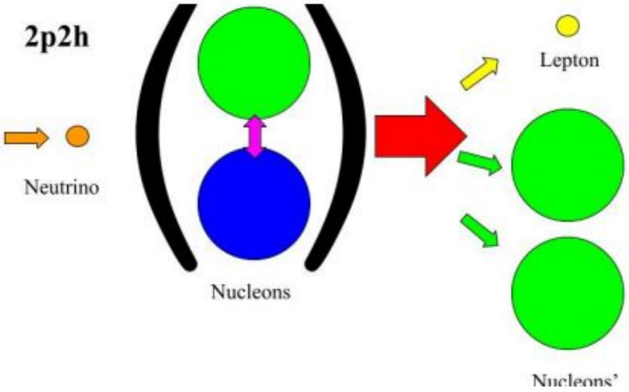




2 particle, 2 hole (2p-2h)

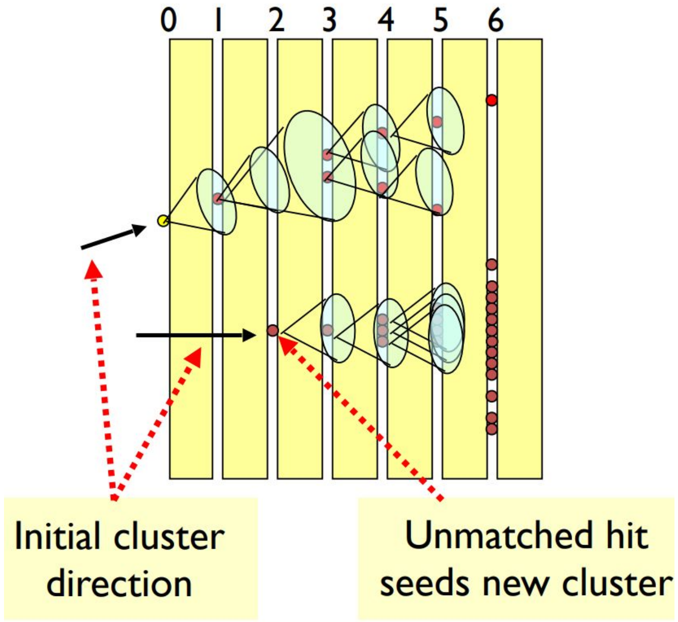
- Strongly correlated nucleon pairs are both scattered
- Generally lower cross section vs other CC channels





Reco Methods (Pandora)

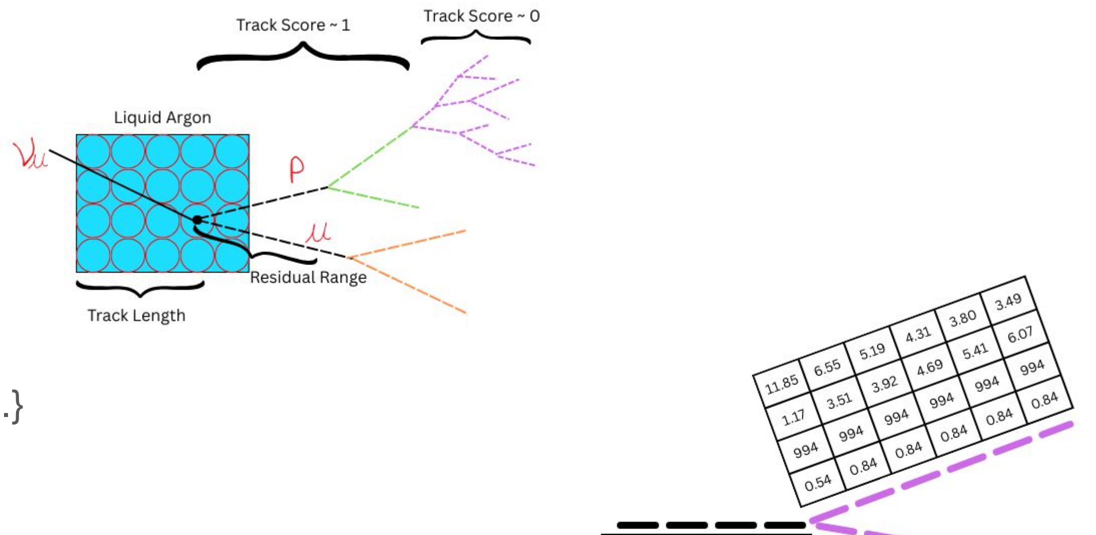
- ~130 different algorithms, 3 ML instances
- From detector (wires): drift time, charge, wireID
- Per plane- 2D clustering
- Combine the three planes- Candidate tracks
- Determine what created the track- beam or cosmic (BDT's)
- Find the vertex (BDT's)
- Assign a track score (BDT's)



Limitations? No particle identification

Data

- Event 1 {
  - Track 1 {
    - TrackLength{…}
    - ResidualRange {...}
    - dE/dx{...}
    - TrackScore{...} } }
  - Track 2 { …



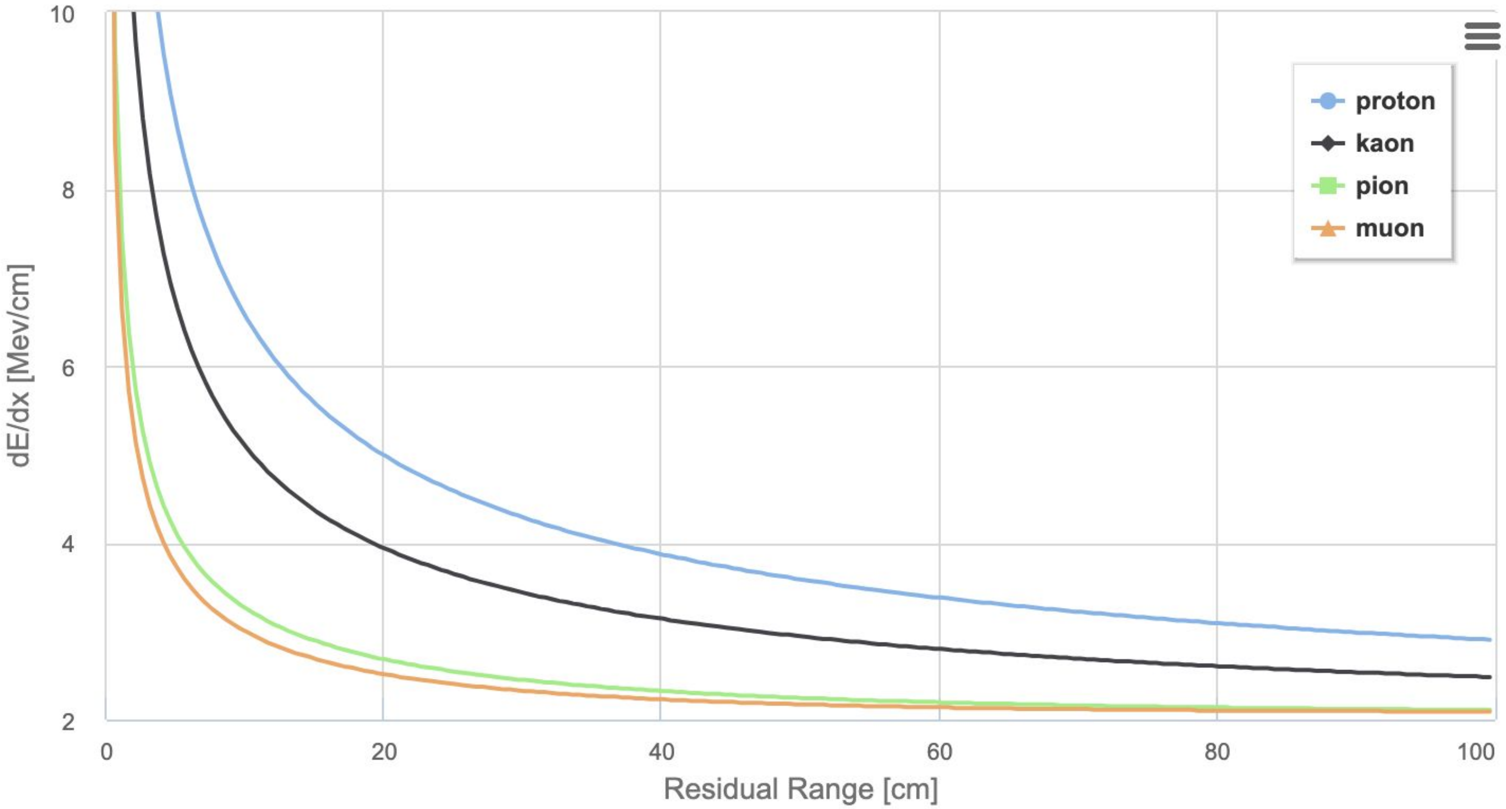
dE/dx	17.53	49.88	2.83	2.90
Residual range	0.27	0.83	3.39	4.89
Track Length	6.48	6.48	6.48	6.48
Track Score	0.54	0.54	0.54	0.54

- dE/dx: How much energy was lost in each step of the particle
- Residual Range: Distance from original neutrino-neutron interaction
- TrackLength: How far neutrino traverses argon before interaction (same)
- TrackScore: How likely this track is to be a particle or a shower

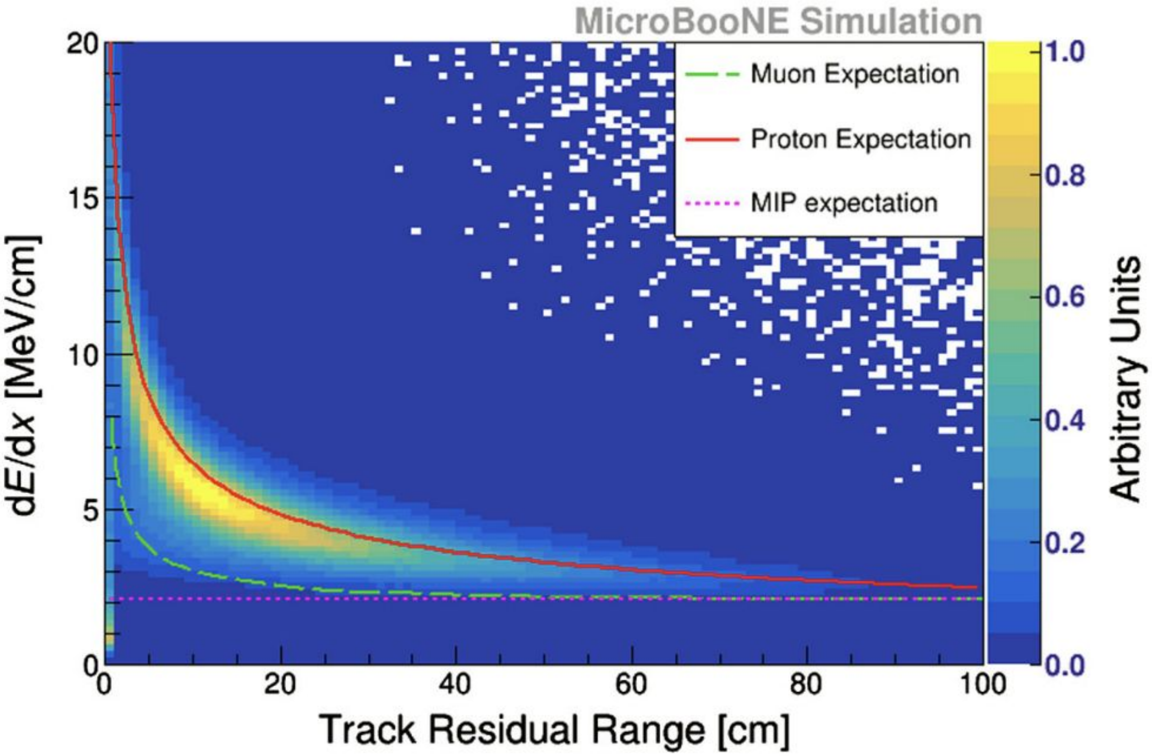
Particle Identification

- Particles lose energy in Argon in distinct ways that help identify the particle
  - Electrons and photons form showers
  - Muons, charged pions, protons, and kaons moves in straight lines losing energy according to the Bethe-Bloch formula
  - Particles that stop in the detector will produce a characteristic Bragg peak
- To this end, we can graph dE/dx and Residual Range to find a characteristic energy loss profile
- DUNE interactions will mostly produce protons, pions, and muons

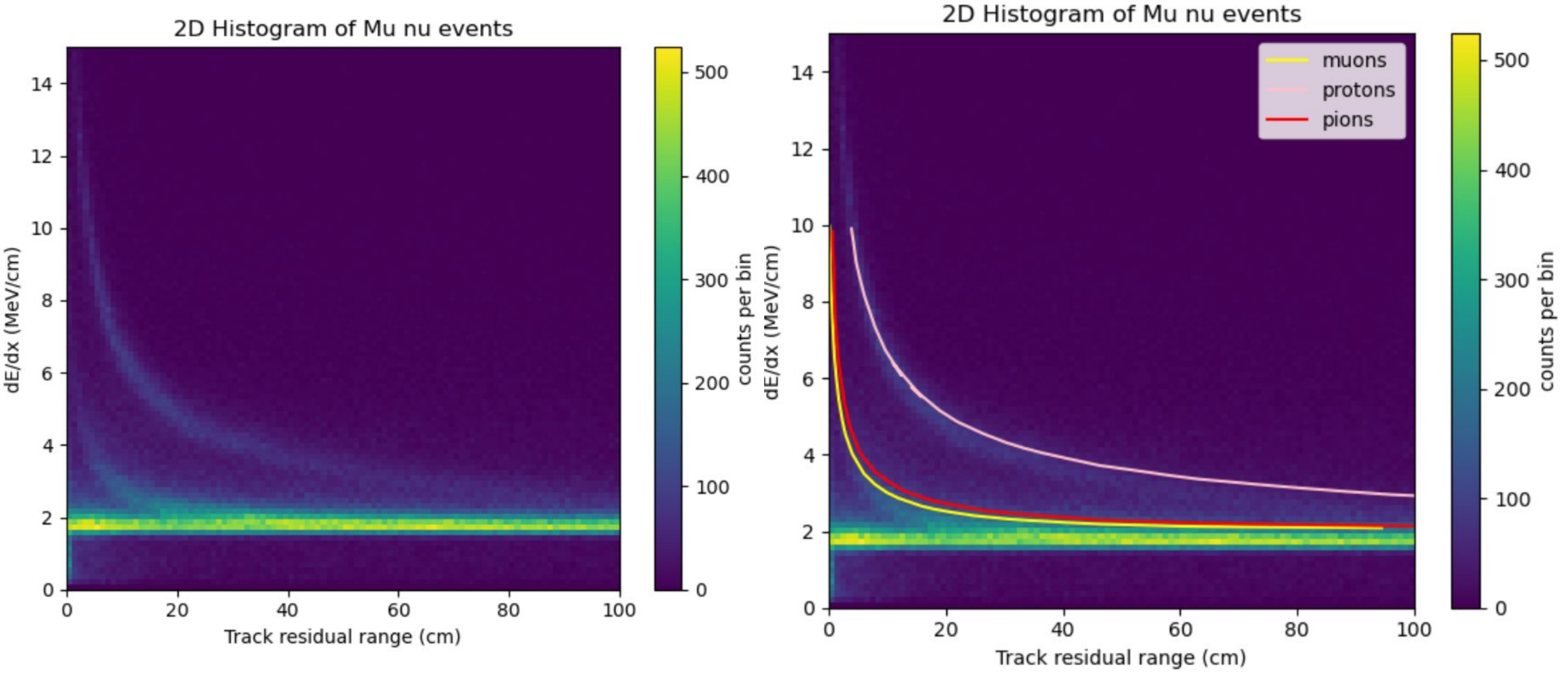
Particle Identification - Expected Curves



Particle Identification - MicroBoone Simulation



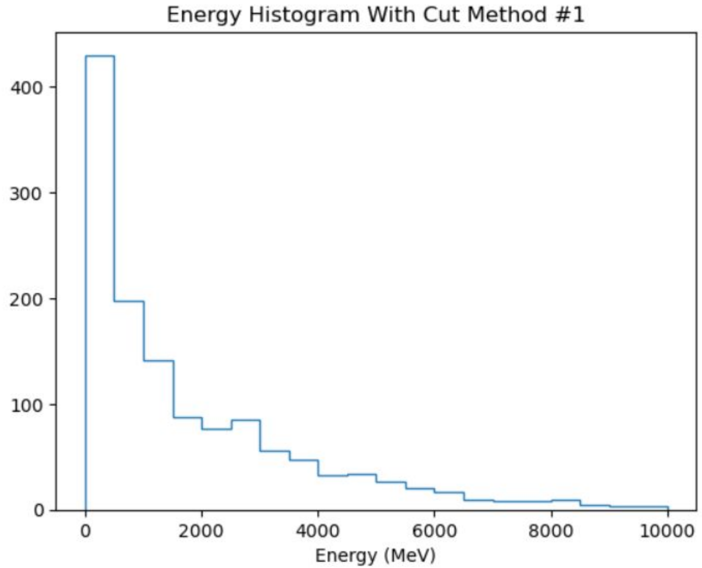
Particle Identification Comparisons



Two Different Methods of Cuts

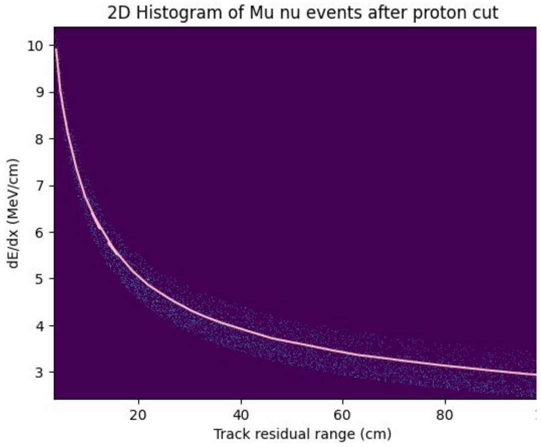
```
[287]: def protoncut(reco):
           mask_fast = (reco.recodEdx >= 7.5) & (reco.recoResidualRange< 10) & (reco.recoTrackScore >= 0.70)
           mask_chill = (reco.recodEdx >= 4) & ( 10 < reco.recoResidualRange)& (reco.recoResidualRange < 20) & (reco.recoTrackScore >= 0.70)
           mask_slow = (reco.recodEdx > 3.7) & (21 < reco.recoResidualRange) & (reco.recoTrackScore >= 0.70)
           protons_maybe = reco[(mask_fast) | (mask_slow)| (mask_chill)]

           return protons_maybe
       protons_maybe = protoncut(rec)
       #get rid of empty particles
       mask_1 = ak.num(protons_maybe, axis =2) > 0
       protons2 = protons_maybe[mask_1]
       #get rid of emptry events
       mask_3 = ak.num(protons2, axis =1) >0
       protons3 = protons2[mask_3]
```

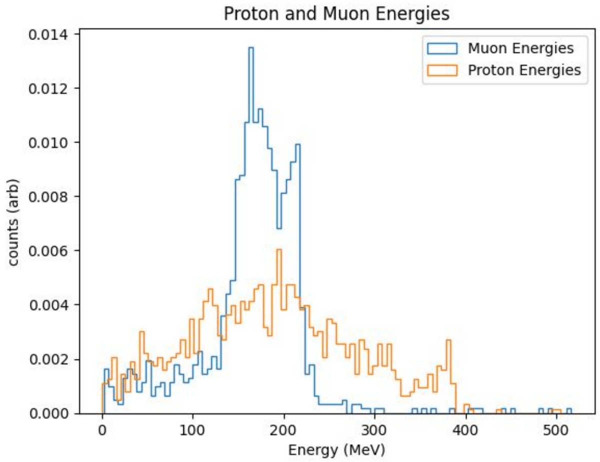
- 1381 Protons
- E = ∑(de/dx) *dx
- Layne figured out a better way to do it, not need to fix this method



2nd Method:

- 1250 Protons





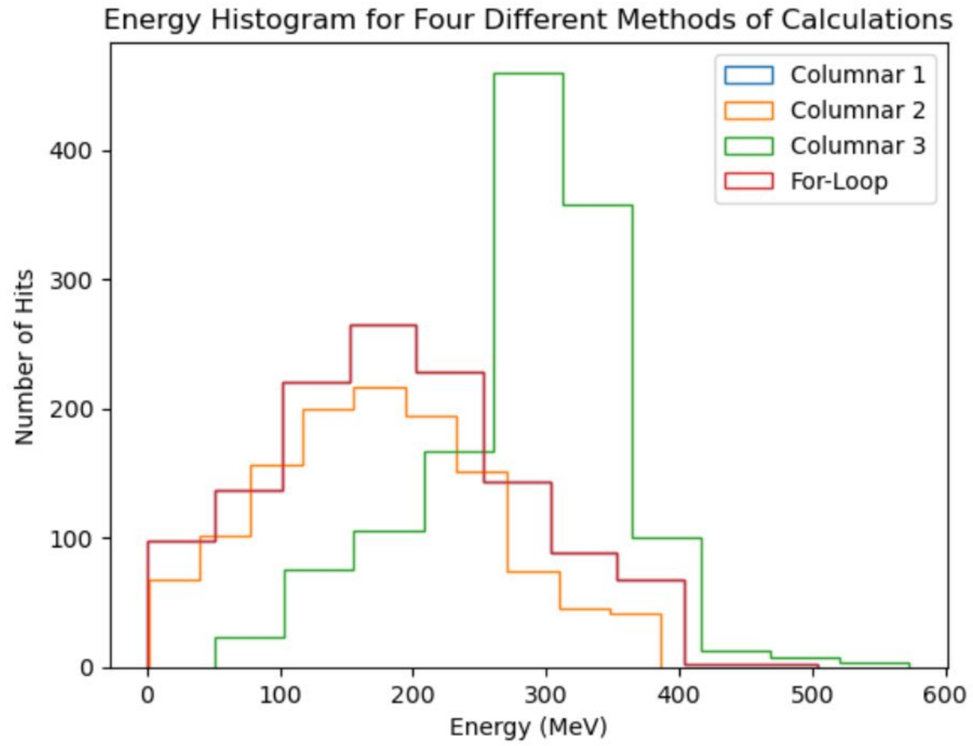

```
filtered_dedx = ak.flatten(rec.recodEdx[cut],axis=-2)
filtered_range = ak.flatten(rec.recoResidualRange[cut],axis=-2)
filtered_length = ak.flatten(rec.recoTrackLength[cut],axis=-2)

filtered_points = ak.zip({"x": filtered_range, "y": filtered_dedx})

newmuon_points = np.stack((new_muonrange, new_muondedx), axis=1)
newproton_points = np.stack((new_protonrange, new_protondedx), axis=1)

muontree = cKDTree(newmuon_points)
protontree = cKDTree(newproton_points)

flat_range = ak.to_numpy(ak.ravel(filtered_range))
flat_dedx = ak.to_numpy(ak.ravel(filtered_dedx))
flat_points = np.column_stack((flat_range, flat_dedx))

muondistances, _ = muontree.query(flat_points)
protondistances, _ = protontree.query(flat_points)

counts = ak.num(filtered_range)

print(ak.sum(counts))

muon_cut = ak.unflatten(muondistances < threshold, counts)
proton_cut = ak.unflatten(protondistances < threshold, counts)

filteredmuon_range = filtered_range[muon_cut]
filteredmuon_dedx = filtered_dedx[muon_cut]
filteredmuon_tracklength = filtered_length[muon_cut]

filteredproton_range = filtered_range[proton_cut]
filteredproton_dedx = filtered_dedx[proton_cut]
filteredproton_tracklength = filtered_length[proton_cut]
```

Energy Distributions for Protons:

Three Different Calculations:

1. (RR(N) - RR(N-1))*Avg(dE/dx) = A
   E = ∑A
2. (RR(N) - RR(N-1))*dE/dx(N) = B
   E=∑B
3. RR[:-1]* ∑(dE/dx) = E

For-loop used #1, cannot see if on graph, they're the same woo hoo



Timing With Columnar Analysis

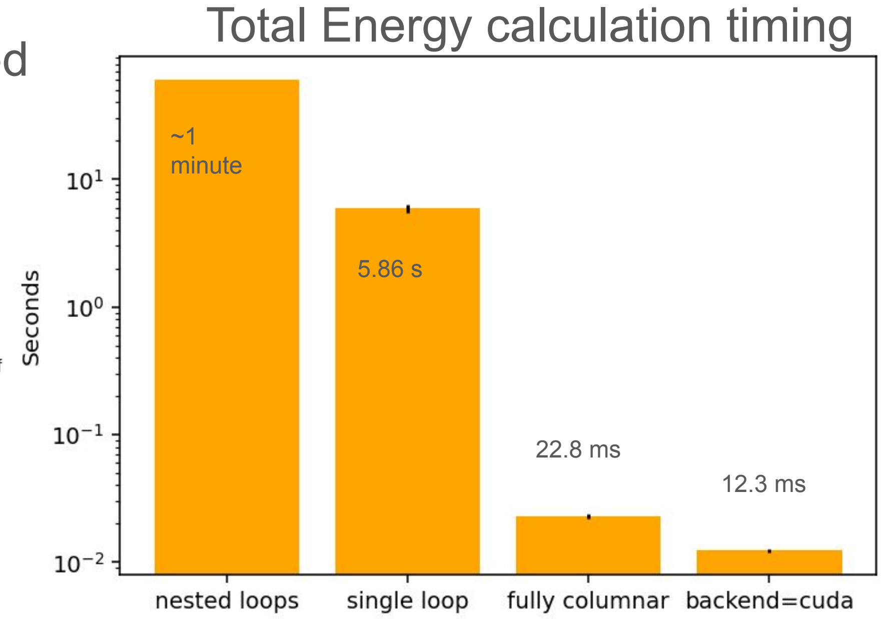
- Only 1,000 events
- GPU tradeoff is still positive

- Fully row-based calculation was killed on METIS login

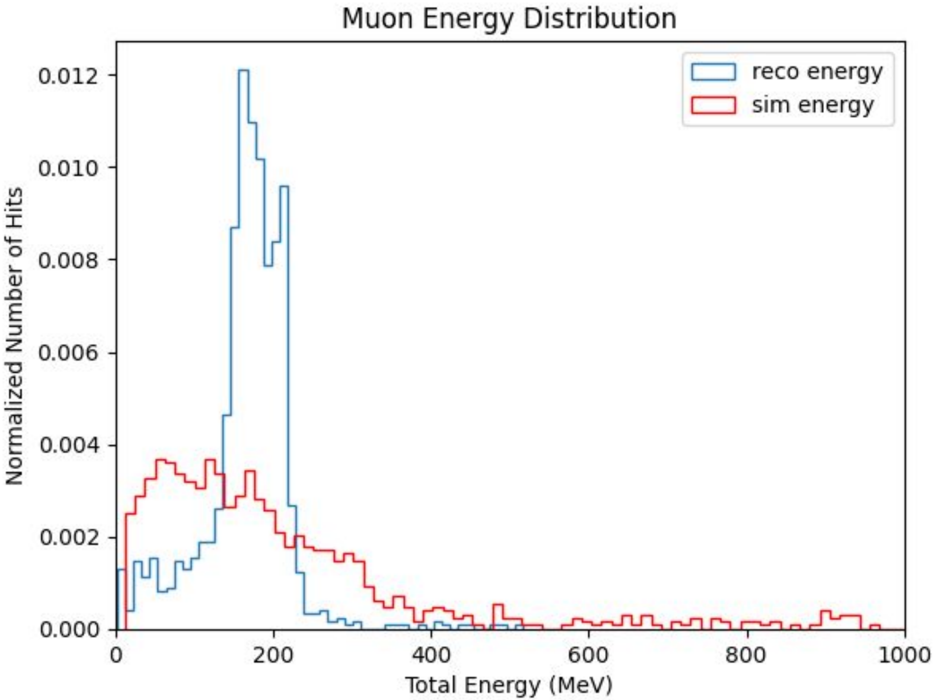
One process has been killed due to excessive CPU usage for more than 30:00:

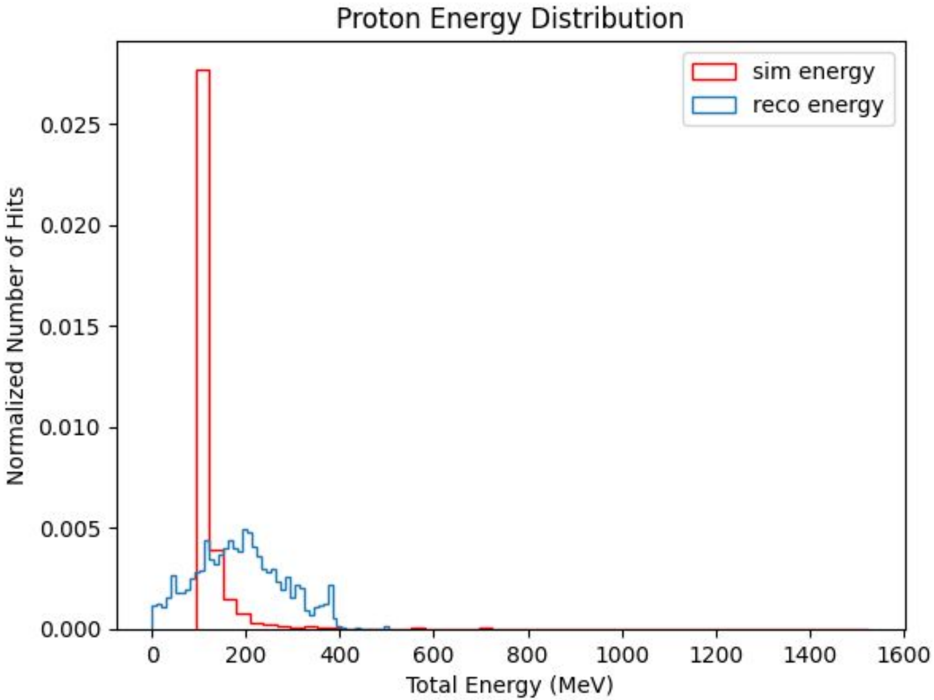
Time:	Apr 27 14:10:54
Node:	metis
PIDs:	154428
Parent PID:	125799
User:	z2039949
Command:	/home/z2039949/.conda/envs/columnar_env/bin/python -Xfrozen_modules=off .local/share/jupyter/runtime/kernel-ab39a4c8-72ac-4bd3-bdab-888ef111f5b5.json
CPU %:	42.22%
Memory %:	0.15%
Run time:	30:01

High-load processes are expected to be run on a compute node rather than a login node. Please consult the NIU CRCD policy page for more information:



Compare to Sim





Conclusion

- Proto-Dune/ Dune produces a lot of raw data, need columnar analysis
- Developed cuts for particle identification
- Compared three different calculation methods
- Discovered the columnar analysis is in fact extremely useful
- Tried to compare to sim

Future Work:

- Reconcile differences between reco and sim
- Larger sample size
- Tau neutrino events
- Reconstruct neutrino beam energies